

- (1) Let $\varphi : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a holomorphic map, and φ_j its j -th coordinate function.
- (a) Show that $\varphi^* dz_j = \partial \varphi_j$ and $\varphi^* d\bar{z}_j = \bar{\partial} \bar{\varphi}_j$. Hence conclude that φ^* maps $\Omega^{p,q}$ into itself.
- (b) Prove that φ^* commutes with both ∂ and $\bar{\partial}$. (Hint: Use the relation $\varphi^* d = d\varphi^*$.)
- (2) Let M be a complex manifold.
- (a) Show that, for any smooth function $f \in C^\infty(M, \mathbb{C})$,

$$\partial \bar{f} = \overline{\bar{\partial} f}.$$

Hence f is holomorphic if and only if $\bar{f}$ is anti-holomorphic.

- (b) A form $\omega \in \Omega^{p,q}$ can be locally expressed as $\sum a_{I,J} dz_I \wedge d\bar{z}_J$, where I, J are multi-indices with $|I| = p$ and $|J| = q$ and $a_{I,J} \in C^\infty(U, \mathbb{C})$ defined on a coordinate neighbourhood U . Define

$$\tau(\omega) = \sum \overline{a_{I,J}} d\bar{z}_I \wedge dz_J.$$

Prove that τ defines an $\mathbb{R}$ -isomorphism $\Omega^{p,q} \rightarrow \Omega^{q,p}$.

- (c) Show that $\partial \circ \tau = \tau \circ \bar{\partial}$. Hence prove that the cohomology of the cochain complex $(\Omega^{*,*}, \bar{\partial})$ is isomorphic to $(\Omega^{*,*}, \partial)$.

- (3) Let $z_j = x_j + i y_j$, $j = 1, \dots, n$ denote the complex coordinates on $\mathbb{C}^{2n}$. Show that

$$\frac{i}{2} \sum_j dz_j \wedge d\bar{z}_j = \sum_j dx_j \wedge dy_j.$$

- (4) Let (M, ω, J) be a Kahler manifold. Recall that, considering ω as a complex valued form we can express it as

$$\omega = \frac{i}{2} \sum h_{jk} dz_j \wedge d\bar{z}_k,$$

where (h_{jk}) is a positive definite hermitian matrix.

- (a) Prove that $\omega(\frac{\partial}{\partial x_p}, \frac{\partial}{\partial x_q}) = -\text{Im } h_{pq} = \omega(\frac{\partial}{\partial x_p}, \frac{\partial}{\partial x_q})$.
- (b) If (ω, J, g) is a compatible triple then $g(\frac{\partial}{\partial x_p}, \frac{\partial}{\partial x_q}) = \text{Re } h_{pq} = g(\frac{\partial}{\partial x_p}, \frac{\partial}{\partial x_q})$.
Therefore,

$$\begin{aligned} h_{pq} &= \text{Re } h_{pq} + i \text{Im } h_{pq} \\ &= g(\frac{\partial}{\partial x_p}, \frac{\partial}{\partial x_q}) - i \omega(\frac{\partial}{\partial x_p}, \frac{\partial}{\partial x_q}) \end{aligned}$$

(c) Define $h = g - i\omega$. Show that h is a hermitian metric on (TM, J) (meaning that $h(Ju, v) = ih(u, v)$, $h(u, Jv) = -ih(u, v)$ etc).

(d) Prove that h is positive definite. (In fact, h is positive definite if and only if (h_{pq}) is positive definite.

(5) Let (V, J) be a vector space with a complex structure J , and let g be an inner-product on V which is J -invariant. Then we can extend g to a complex bilinear map H on $V \otimes \mathbb{C}$ as follows:

$$H(u \otimes \lambda, v \otimes \mu) = \lambda \mu g(u, v).$$

(Note that it is not a complex inner product.)

(a) Show that H vanishes on $V_{1,0} \otimes V_{1,0}$ and $V_{1,0} \otimes V_{1,0}$.

(b) Define $h : V \times V \rightarrow \mathbb{C}$ by

$$h(u, v) = 2H(u_{1,0}, v_{0,1}),$$

where $u_{1,0} = \frac{1}{2}(u \otimes 1 - Ju \otimes i)$, and $v_{0,1} = \frac{1}{2}(v \otimes 1 + Jv \otimes i)$.

Prove that $h(Ju, v) = ih(u, v)$ and $h(u, Jv) = -ih(u, v)$. Thus h is a Hermitian inner product on (V, J) .

(c) Prove that $h = g - i\omega$, where ω is a symplectic form on V . In fact, (ω, J, g) is a compatible triple.

(d) Let g be the canonical inner product on $\mathbb{R}^{2n}$ and J the canonical almost complex structure. Determine H , h and ω following the above construction.

(6) Let (M, J) be an almost complex manifold and g be a Riemannian metric on M which is J invariant. Define a 2-form ω by $\omega(u, v) = g(Ju, v)$ for all $u, v \in T_p M$, $p \in M$. Then the following are equivalent:

- J is integrable and ω is closed;
- (M, ω, J) is Kahler.